

M002 Week 4 — English Worksheet

Topic: Midpoint Check — Build a Maze & Solve It · **Course:** Maze Madness · **Time:** about 45 minutes

This is your **midpoint check**. You design a maze yourself, then solve it with the code you learned in Weeks 1–3.

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
keep doing forever:
  if wall ahead:
    turn right
  otherwise:
    move forward
```

This is a classic "right-hand wall follower". On an L-shaped maze, does the agent reach the goal? Why or why not?


```
keep doing forever:
  if no wall ahead:
    move forward
  otherwise if no wall on right:
    turn right
  otherwise:
    turn left
```

Two things can go wrong with this agent on your own maze. Name one of them.

A maze with:

- 3 corners
- 1 dead end
- 1 path to the goal

Sketch a quick floor plan of how this maze could look. (Draw boxes for walls.)

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — At a dead end (walls on the front AND right AND left), the agent should **turn around** (turn right twice).

```
if wall ahead AND wall on right AND wall on left:  
    turn right
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — At a fork, the agent should **prefer going forward** if it can, then right, then left.

```
if no wall on left:  
    turn left  
otherwise if no wall on right:  
    turn right  
otherwise:  
    move forward
```

Hint: the priorities are upside-down.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The agent's loop never stops, even after reaching the goal.

```
keep doing forever:
  if wall ahead:
    turn right
  otherwise:
    move forward
```

Hint: swap "keep doing forever" for a while loop that stops at the goal.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. **Design and build your own 2D maze.** It must include:

- at least **3 corners**
- at least **1 dead end**
- a clear **start** and **goal**

Then solve it with your own agent code (while loop + conditions). When you finish, come back here.

Send a photo or video of the agent solving your maze, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I sketched the maze with ...

My maze has ... corners and ... dead ends.

My agent's strategy was ...

The agent got stuck when ...

I fixed it by ...

If I had more time, I would add ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while the agent runs your maze. Talk through it like you are teaching someone who has never seen it. Try to use these words: **maze, corner, dead end, strategy, stuck**.

1. Show the maze from above and point out the start, the goal, and the dead end.
2. Run your code and follow the agent through.
3. Show one bug you hit and how you fixed it.
4. Say what your agent's strategy is in one sentence.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.